Scientific Importance

The flux line lattice (FLL) in type II superconductors is a remarkably tunable form of emergent spatial order in solid-state materials. Unlike other correlated states such as charge or spin density waves, the wavelength of the FFL is *continuously* tunable with magnetic field H and temperature T within the superconducting state. In some cases, the lattice symmetry can also be controlled by H and T [1]. This tunability of the FFL has made it a fruitful platform to study structural phase transitions of interacting vortices within a superfluid background, and small angle neutron scattering (SANS) has been instrumental to mapping its spatial structure.

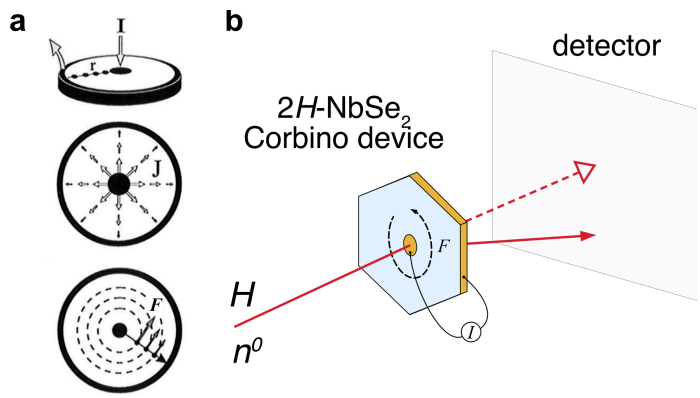

Figure 1: ***a*** *The Corbino geometry exhibits spatially varying current density J, resulting in a transverse force gradient. Adapted from [2].* ***b*** *Schematic Corbino device. We propose to induce rotation of the FLL with J and measure the structure in situ with SANS utilizing our newly-developed symmetry-adapted filters [3].*

Beyond equilibrium, the FLL dynamics are similarly, if not more, interesting. However, system dynamics are difficult to directly study because ubiquitous diffraction methods cannot capture translational motion. To overcome this challenge, team members recently studied a skyrmion lattice (SkL) system by applying a driving force to induce *rotational* motion. Then, using advanced ML-inspired algorithms, SkL dynamics were automatically extracted from SANS data. Here we would like to extend this approach to study dynamics of a canonical flux-lattice material, $2H$-$NbSe_2$. The approach uses a Corbino geometry (Fig. 1a) and electrical current (or in the case of the SkL, a heat current) to drive the azimuthal motion of the FLL. As well as fundamental insights into factors that affect FLL dynamics, the approach once demonstrated could be used in the future to study flux pinning and flux dynamics in the presence of nanostructuring and other heterogeneities, topics that are of considerable technological interest.

This particular experiment is motivated by transport experiments that indirectly identify an FLL structural phase transition as a function of drive current density J_d [2]. We aim to explore two key phenomena directly with SANS. *(1)* The variation of J_d along the radial direction creates a corresponding variation in the transverse force on the FLL (Fig. 1a). As a result, we expect to observe shear distortions in the FLL, which could allow us to extract the FLL shear modulus for the first time (as far as we are aware). *(2)* Beyond shear distortion, SANS imaging of driven SkLs indicates that the precise balance of J_d, pinning energy, and vortex density can tip the balance between global shear distortions or local rotation of the lattice [3]. Understanding these effects will provide new insights into the elastic properties, pinning effects, and non-equilibrium behavior of vortex matter, paving the way for future studies wherein we lithographically pattern pinning sites to control vortex dynamics.

Preliminary Work

Manipulation of SkLs using a radial electric current in a Corbino geometry was successfully studied by team members using SANS [4] (see also https://tinyurl.com/mt5wjukj). To facilitate rapid and effective data analysis, a specialized algorithm was constructed that extracts the orientation angles of underlying lattices that exhibit certain rotational symmetries. The approach applies a symmetry-adapted filter (Fig. 2) to each diffraction pattern and maximizes an overlap score (the inner product of the filter and the image) by rotating the filter in a regression loop. This method was successful in extracting angular velocities from thousands of diffraction images, and will be detailed in an upcoming publication. Additionally, we have expertise in synthesizing high-quality TMD superconductors (essential for studying high-quality FLL behavior [7,8]) and in the technical requirements for the Corbino measurements. Building on this foundation, we are well-positioned to study rotational dynamics of FLLs in $2H$-$NbSe_2$ using SANS at HFIR.

Choice of Instrument

GP-SANS is the most suitable instrument for this proposal as it achieves the lowest q among the SANS instruments at ORNL while accommodating high-field, low-temperature sample environments. Additionally, it offers precise sample rotation, enabling *in situ* tuning of the measurement geometry to achieve optimal diffraction conditions.

Experiment Plan

A crucial aspect of SANS with FLLs is collecting sufficient scattered counts. To maximize the signal, following prior SANS efforts [5], we will grow large single crystals reaching $10 \times 10 \times 1$ mm^3 in size with a modified vapor transport technique [6]. This lateral size is well-suited to the typical SANS beamspot at HFIR.

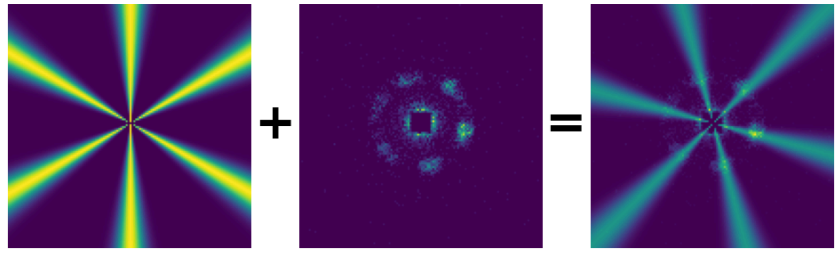

Figure 2: Symmetry-adapted filter algorithm applied to time-series SANS data in Cu_2OSeO_3, allowing for rapid lattice orientation extraction.

Our team also has extensive experience with focused ion beam (FIB) micro-sculpting [9,10], which we will use to fabricate $2H$-$NbSe_2$ Corbino devices mounted on ~500 nm thick amorphous Si_3N_4 membranes (commercially available); Fig. 1b shows a schematic of the device and measurement geometry. By using FIB as opposed to hand fabricating the devices, *(1)* we can "drill" through the crystal and deposit platinum in ultra-high vacuum to create spatially symmetric contacts and eliminate contact degradation by surface oxides. *(2)* The nitride membrane contribute minimal background and, moreover, can be sculpted into spring-like geometries to relieve thermal strain [7]. We will pre-screen the devices after fabrication to ensure they are robust against thermal cycling, and can sustain the J_d required to induce FLL dynamics.

Previous transport measurements on $2H$-$NbSe_2$ show critical de-pinning current densities spanning $J_p \sim 10^1 - 10^2$ A/cm^2, where the lower limit is realized in "cleaner" samples [8]. For a cross-section 10×1 mm^2, this would imply current in the range 1 A – 10 A to induce FLL motion. Technically, this magnitude of current is straightforward to control, however, low contact resistance is crucial to minimize Joule heating; FIB techniques will be especially useful in this context. We will work in the regime $J_d \gtrsim J_p$ such that the FLL rotation is slow enough for data collection. In the event the rotation is too fast to resolve, we envision AC driving current with a lock-in detection scheme could allow us to extract FLL time dependence.

We will start our experiment by roughly confirming with transport the temperature-field phase diagram measured at Columbia to (1) align the sample and (2) find the FLL signal. We will optimize signal strength vs. measurement rate and estimate the FLL rotation velocity we can tolerate for the dynamic measurements. We estimate this will take one day. On the subsequent day, we will determine J_p with transport measurement while monitoring the SANS signal; preliminary J_p estimates from transport measurements at Columbia will narrow the search window. We will also check that Joule heating is minimal by ensuring that current-voltage curves are reversible. After completing these calibration procedures, we will drive the sample at $J_d \gtrsim J_p$, and monitor for FLL depinning and motion. We will adjust J_d to generate FLL motion that is slow enough to be captured by the optimal measurement parameters determined earlier. On the last day, we aim to increase J_d into the large drive limit beyond the J_p limit to look for novel, dynamic FLL phase transitions that are anticipated (Fig. 1a). In total, we estimate needed (3) days total to perform this experiment; the team brings extensive experience performing these experiments [9].

Data Analysis and Scientific Outcomes

The rotational dynamics of the FLL will be analyzed with the aforementioned symmetry-adapted filter technique. This physics-informed approach will allow for rapid analysis of rotational velocity. With this algorithm, large datasets can be rapidly analyzed, revealing effects of pinning sites, symmetry changes/breaking, and trends in FLL dynamics. In analogy with temporal lock-in methods, the SAFs act like an angular lock-in approach that can allow weak symmetry-respecting signals to be extracted

Safety Considerations

For this experiment, we will adhere to the guidance of the subject matter expert on the cryomagnet, as well as follow all safety protocols related to handling cryogens.

References

[1] A. Huxley *et al.*, Nature 406, 160 (2000).
[2] D. López *et al*., Phys. Rev. Lett. **82**, 1277 (1999).
[3] H. Jin *et al.*, Nano Lett. 24, 12226 (2024).
[4] N. Chalus *et al.*, Phys. Rev. B **111**, 064410 (2025).
[5] C. S. Oglesby *et al.*, Journal of Crystal Growth 137, 289 (1994).
[6] P. J. W. Moll, Annual Review of Condensed Matter Physics 9, 147 (2018).
[7] C. Guo *et al.*, Nat. Phys. 20, 579 (2024).
[8] S. Bhattacharya and M. J. Higgins, Phys. Rev. Lett. 70, 2617 (1993).
[9] S. Mühlbauer *et al.*, *Rev. Mod. Phys.* **91**, 015004 (2019).